1. Road Network Flood Prediction (MSc): Developed a logistic regression model using Sentinel-1 SAR data that achieved 74% accuracy in predicting disruptions across 68,404 road segments during the 2018 Kerala floods.
2. Urban Road Resilience Assessment (BTech): Quantified urban road resilience using a response curve approach, demonstrating that traffic rerouting can reduce flood-induced community disruption by up to 94.04%.
3. Sasthamcotta Lake Shrinkage Analysis: Performed multi-temporal NDWI analysis on Landsat 8 imagery to reveal a 41.31% reduction in surface area between 2002 and 2017.
4. Ecological Modelling: Scottish Pine Martens: Utilized cost-surface modeling and graph theory in R to map vital wildlife corridors and 330 critical conifer "stepping stones" for population recovery in Scotland.
5. Urban Greenspace Accessibility Mapping: Benchmarked accessibility against national standards to reveal that only 13% of Greater Manchester's population can access a 100-hectare greenspace within 5 km.
6. Python: Weighted Redistribution Algorithm: Implemented a Python tool to correct centroid bias for 1,023 tweets by redistributing points based on population density and likelihood regions.
7. Passu Glacier Runoff Modelling: Simulated glacial runoff dynamics by calibrating temperature-driven melt parameters to achieve a precise "best fit" with observed peak flows of 78.1 m^3/s

Machine Learning Model for Predicting Road Network Disruptions During Heavy Precipitation: A Case Study on the Kerala Floods 2018

My MSc dissertation, titled **"Machine Learning Model for Predicting Road Network Disruptions During Heavy Precipitation: A Case Study on the Kerala Floods 2018,"** represents a sophisticated geospatial approach to enhancing urban resilience. Below is a deep explanation of the project, from its fundamental motivations to its final quantitative conclusions.

1. The Basics: Motivation and Study Context

The research was driven by the global trend of intensifying flooding events, exacerbated by climate change and human modifications to natural drainage systems. **Road networks** are identified as critical urban lifelines; when they fail during a flood, emergency response is crippled and communities are isolated. You focused on the **2018 Kerala floods**, the state's most catastrophic event since 1924, which caused extensive damage to road infrastructure and rendered many routes impassable. The goal was to move beyond traditional hydrological models—which often oversimplify complex terrain—and use **machine learning** to predict specific road segment vulnerability.

2. Data Acquisition: Types, Sources, and Rationales

Because historical data on specific road flooding was scarce, you generated a primary dataset by integrating multiple remote sensing and geospatial sources:

Data Type	Source	Rationale for Use
Road Network	**Geofabrik (OpenStreetMap)**	To obtain detailed shapefiles and unique IDs (osm_id) for all **68,404 road segments** in Ernakulam and Idukki.
Flood Detection	**Sentinel-1 SAR (ESA)**	**Crucial Decision:** Optical imagery (like Landsat) was 100% obscured by clouds during the 2018 floods. **Synthetic Aperture Radar (SAR)** was used because it penetrates clouds and rain to accurately map water pixels.
Elevation & Slope	**SRTM DEM (NASA/NIMA)**	A 30m Digital Elevation Model was used to extract topography, as lower elevations and specific slope gradients direct water flow.
Precipitation	**CHIRPS**	Provided high-resolution (~6km) satellite-based rainfall estimates to account for spatial variability in precipitation across the districts.
Imperviousness	**Tsinghua University (GAIA)**	measuring surface permeability is vital; urban concrete/asphalt prevents absorption and increases surface runoff.
Drainage Network	**OpenStreetMap**	Used to calculate the distance of each road to the nearest drainage source, reflecting inherent flood risk.

3. Technical Methodology and Key Decisions

The project followed a systematic workflow involving complex data processing and algorithmic refinement:

- **Feature Extraction:** Using **ArcGIS Pro**, I extracted 11 environmental features for every road segment, including mean/min/max values for elevation and slope, precipitation intensity, and distance to drainage.
- **Handling Data Imbalance (The SMOTE Decision):** A major technical hurdle was that only **1.12% (765 segments)** of the roads were actually flooded. To prevent the model from simply "guessing" that no roads would flood, I applied the **SMOTE (Synthetic Minority Over-sampling Technique)** algorithm. This created synthetic examples of flooded roads to create a balanced training set, ensuring the model learned the characteristics of the minority class.
- **Feature Selection:** I used **Recursive Feature Elimination (RFE)** and Logistic Regression p-values to prune insignificant variables. This narrowed the model down to the **five most influential**

predictors: mean slope, mean precipitation, imperviousness, minimum slope, and maximum slope.

- **Model Selection:** I chose **Logistic Regression** for its efficiency in binary classification and its high interpretability, allowing you to see exactly how much each factor (like a unit increase in slope) affected the log-odds of flooding.

4. Results and Performance Metrics

The model's performance was rigorously tested and validated:

- **Accuracy:** The final model achieved an **overall accuracy of 74%**.
- **Classification Metrics:** For the "Flooded" class, the model showed a **precision of 0.78**, meaning it was highly reliable when it predicted a flood.
- **Reliability:** I applied **k-fold cross-validation** (k=10), which yielded a consistent accuracy of **0.7367 (+/- 0.0046)**, proving the model is stable and not overfitted to a specific data subset.
- **Predictive Power:** The **Area Under the Curve (AUC)** was **0.74**, indicating good discrimination between flooded and non-flooded segments.

5. Conclusions and Practical Significance

The research proved that **urbanization (imperviousness)** significantly boosts flood likelihood, while roads at **higher minimum elevations** are safer. The final conclusion highlights that this data-driven approach empowers authorities to move from "reactive" to **"proactive" disaster management**. By predicting disruptions in advance, planners can **reroute traffic**, reinforce vulnerable infrastructure, and deploy emergency teams more effectively, ultimately fortifying community resilience against nature's unpredictable wrath

Assessment of Resilience of Road Networks to Flooding in Kozhikode Municipal Corporation

For my BTech final project at the National Institute of Technology Calicut, I conducted a study titled **"Assessment of Resilience of Road Networks to Flooding in Kozhikode Municipal Corporation"**. The project was born out of a critical gap in urban planning: while most flood assessments focus on monetary loss, I wanted to quantify the actual disruption to human life and the ability of a city to recover and adapt to such shocks.

The Basics: Motivation and Study Area

I focused on the **Kozhikode Municipal Corporation** in Kerala, which is a major transportation hub for the state. After the catastrophic floods of 2018 and 2019, it became clear that our infrastructure needed to be more than just "strong"—it needed to be **resilient**. I defined resilience as the ability of a system to anticipate, absorb, adapt, and recover from disturbances. My goal was to move beyond traditional hazard mitigation to a system where road networks maintain their functionality even during extreme events.

Data Acquisition: Types, Sources, and Rationales

To build a comprehensive model, I had to integrate a wide variety of data types, many of which required manual digitization or extraction due to the lack of a proper existing database:

Data Type	**Source**	**Rationale for Use**
Administrative Boundary	Dept. of Town & Country Planning	I used a 2011 Land Use Map as a base to georeference and digitize the corporation boundary in **QGIS**.
Transportation Network	**OpenStreetMap (OSM)**	I imported detailed vector layers for highways (primary to service roads), railways, and stations to map the city's lifelines.
Drainage System	OSM	Crucial for identifying how floodwater is removed from the city.
Flood Prone Maps	Dept. of Town & Country Planning	Used to identify historical floodplains and validate which roads were most vulnerable.
Population Data	**2011 Census**	Vital for calculating my primary metric: **Disruption** in terms of "person x days".
Hydrological Data	Kuniyil River Flow Data (1979–2020)	I used 40 years of discharge data to calculate the **Annual Exceedance Probability (AEP)**.

The Methodology: Key Decisions and Metrics

I chose a two-step approach: investigating past responses to flood shocks and identifying measures to improve future resilience.

1. **The Disruption Metric:** I calculated disruption as the product of the number of people affected and the delay in their travel time compared to normal conditions ($Disruption = People \times (TravelTime - TypicalTravelTime)$).
2. **AEP Calculation:** Using **Weibull's method**, I ranked the maximum annual discharge of the Kuniyil river to determine that the 2018 and 2019 floods had AEPs of **0.05 (1-in-20 year)** and **0.1 (1-in-10 year)**, respectively.
3. **The Base Case:** I assumed a normal travel speed of **30 kmph** and a flooded speed of **15 kmph**. I sorted 38 affected roads and selected the **top five** with the highest disruption for deep analysis.

Future Resilience: Enhancement Measures

I then modeled how the system would respond if three specific resilience properties were added:

- **Redundancy:** I assumed external measures (like sponge roads or elevated sections) allowed vehicles to move at **20 kmph** during floods.

- **Flexibility:** I modeled the effect of **rerouting** traffic to the shortest non-flooded alternate path using OSM and Google Maps.
- **Robustness:** I modeled structural protections (like barricade walls or glass barriers) that allowed heavy vehicles to move at **20 kmph** while lighter vehicles remained slowed.

Results and Conclusions

By plotting the disruption against the AEP, I created **Response Curves** where the area under the curve represented the Expected Annual Disruption (EADIS).

- **Flexibility** proved to be the most effective overall measure, reducing disruption by an average of **72%**. For the **Puthiyangadi-Kakkodi Road**, it achieved a massive **94.04% reduction**.
- However, for the **Kakkodi-Kannadikkal Road**, **Robustness** was superior, offering a **65% reduction** compared to only 36% for flexibility.

I concluded that while rerouting (flexibility) is generally the best way to maintain community connectivity, certain terrains require structural interventions (robustness). This data-driven approach proves that resilience can be quantified, allowing city planners to prioritize interventions that ensure our lifelines remain open when they are needed most.

Spatio-temporal Analysis of the Largest Fresh Water Lake in South West India

For my project, **"Spatio-temporal Analysis of the Largest Fresh Water Lake in South West India,"** I conducted a decade-long monitoring study of **Sasthamcotta Lake** in Kerala. My goal was to quantitatively evaluate the environmental degradation and surface area shrinkage of this vital **Ramsar site**, which provides drinking water to over half a million people.

1. The Basics: Motivation and Study Area

Freshwater makes up only 0.01% of the world's water, making its conservation critical. Sasthamcotta Lake is unique because its water is naturally purified by a larval species called *cavaborus*, but it has faced severe stress due to rising temperatures and human intervention. I focused on the period between **2013 and 2023** to understand how the lake's water surface area responded to these pressures. The study area is situated at an elevation of 33m and has a catchment of 934.56 hectares.

2. Data Acquisition: Types, Sources, and Rationales

To perform this analysis, I integrated satellite imagery and historical baseline data:

Data Type	Source	Rationale for Use
Landsat 8 Surface Reflectance	**Google Earth Engine (GEE)**	I chose this dataset for its **30m resolution**, which is ideal for a medium-sized lake like Sasthamcotta. I used **Surface Reflectance (Level 2)** data to ensure atmospheric artifacts (scattering and absorption) were corrected, providing consistent values for long-term monitoring.

Historical Baseline (2002)	**Ramsar Sites Information Service**	I used the recorded 2002 area of **3.73 km²** as my benchmark to calculate the total percentage of shrinkage over time.
Water Withdrawal Data	Literature (Shibu and Ayoob, 2021)	I acquired data on the three treatment plants at the lake, which currently pump **58 million litres per day (MLD)**, to explain the qualitative reasons behind the shrinkage.

3. The Methodology and Key Decisions

My analysis relied on the **Normalized Difference Water Index (NDWI)**, which I calculated using the **Green** (Band 3) and **Near-Infrared** (Band 5) wavelengths from Landsat 8.

- **Decision to use NDWI:** I chose NDWI because it increases the characteristics of water features while reducing the low NIR reflectance of water, effectively enhancing pixels containing water. While modified versions exist for urban areas, I used the original index since Sasthamcotta is not in an urban setting.
- **The Seasonal Filter:** I made the critical decision to filter the Landsat collection to include only images from the **annual dry season (January to March)**. This is when the lake is most vulnerable to drying, providing the most stringent test of its resilience.
- **Workflow in GEE and ArcGIS:** I developed a custom GEE script to calculate the median NDWI values for each year and export the best yearly image. I then moved to **ArcGIS Pro**, using the **Raster Calculator** to reclassify pixels and convert the water surface into a **polygon** to calculate the precise area.

4. Quantitative Results and Findings

My analysis revealed significant variability and a general trend of degradation:

- **Extreme Shrinkage:** From the 2002 baseline of 3.73 km², I found that the lake reached its absolute minimum area in **2017 at just 2.19 km²**. This represented a massive **41.31% reduction** in surface area.
- **The 2013 Crisis:** In 2013, the area was also very low (2.25 km²), and records show the water level actually fell **below sea level** due to low rainfall and high withdrawal stress.
- **Recent Recovery:** Interestingly, my data captured a recovery trend. By **2022**, the lake surface area increased to **3.00 km²**. I attributed this to the exceptionally high rainfall and floods Kerala experienced from 2018 onwards, as well as the introduction of alternative water sources that reduced the pumping stress on the lake.

5. Conclusions

I concluded that the lake has suffered an overall **42% reduction** in area compared to 2002 due to the combined effects of climate change and excessive water withdrawal. My study proved that **Remote Sensing and GIS** are highly efficient tools for monitoring such landforms globally. However, I also noted that area alone is not enough; future work must assess changes in the **physical and chemical characteristics** of the water to fully understand the lake's health.

Ecological Corridor Risk Modelling for Pine Martens in Scotland

For my project, **“Ecological Corridor Risk Modelling for Pine Martens in Scotland,”** I conducted a spatial ecology assessment to support the conservation and recovery of the Scottish pine marten (*Martes martes*) population. My primary objective was to evaluate the functional connectivity between two major protected areas: **Cairngorms National Park** and **Loch Lomond National Park**.

1. The Basics: Motivation and Ecological Concepts

Historically, pine martens were widespread across Scotland’s woodlands, but their population plummeted in the 19th and 20th centuries due to deforestation and persecution. While recent reforestation and legal protections have aided their recovery, I recognized that survival depends on more than just isolated habitat patches; it requires **functional connectivity**.

In this project, I distinguished between **structural connectivity** (the physical arrangement of landscape features) and **functional connectivity** (how the animal actually reacts to those features). My goal was to identify the "stepping stones" (intermediate stopover sites) and corridors that facilitate genetic exchange and population resilience between the two parks.

2. Data Acquisition: Types, Sources, and Rationales

To build a realistic movement model in **R**, I integrated several specific geospatial datasets:

Data Type	Source	Rationale for Use
Landcover Data	**Corine Landcover 2018**	I used this raster layer to identify **34 different landcover classes** across the study extent to determine which areas facilitate or hinder movement.
Landscape Resistance Values	**Stringer et al. (2018)**	I relied on established literature to assign "costs" to different landcover types, representing how difficult it is for a marten to travel across a specific pixel.
National Park Boundaries	OS / Scottish Government	I extracted the boundaries of Cairngorms and Loch Lomond to identify the **centroids** that served as the origin and destination points for my movement models.

Conifer Patches	Habitat Shapefiles	I identified **330 conifer patches** within the study area to analyze them as potential "stepping stones" in a network.
Species Dispersal Distance	Stringer et al. (2018)	I used a **mean dispersal distance of 8 km** to weight my network analysis and create a realistic adjacency matrix for the martens.

3. Technical Methodology and Key Decisions

My methodology was divided into three sophisticated phases implemented entirely in **R**:

- **Parameterizing the Cost Layer:** I reclassified the Corine dataset into a resistance map. I made the ecological decision to assign **forests a low resistance (score of 1)** and **urban areas or large waterbodies a high resistance (score of 1000)**.
- **Least-Cost Path Analysis:** I used **Dijkstra's algorithm** (via the accCost function) to generate cumulative cost surfaces originating from the centroids of both parks. I then overlayed these surfaces and made the critical decision to define the **wildlife corridor as the lowest 7% cost layer**, representing the most efficient movement paths.
- **Graph Theory and Network Analysis:** To move beyond simple paths, I modeled the landscape as a network of nodes (conifer patches).
  - **Betweenness Centrality:** I calculated this to find which patches act as vital links in the shortest paths across the network.
  - **Probability of Connectivity (dPC Index):** I calculated the change in connectivity when individual nodes were removed to determine the relative importance of specific patches to the entire ecosystem.

4. Quantitative Results and Findings

- **Landscape Connectivity:** My analysis revealed that the area between the two national parks offers **moderate to low resistance**. This is largely because the terrain is dominated by forests, bushes, and grasslands that favor marten movement.
- **Corridor Identification:** I successfully mapped a potential wildlife corridor connecting the parks through the least-cost surface.
- **Critical Stepping Stones:** My network plots showed that nodes closer to the identified least-cost corridor had significantly **higher betweenness values**. Interestingly, I found that **fewer nodes showed high dPC values**, suggesting that only a small number of specific patches are absolutely critical for maintaining the entire network's connectivity.

5. Conclusions and Reflections

I concluded that while the current landscape supports connectivity, conservation efforts should prioritize the specific "larger nodes" identified in my dPC analysis.

However, I also noted a few limitations: my model relied on literature-based resistance values rather than **observed movement data**. I also used simplified assumptions that did not account for inter-species competition or predation. I proposed that future research should incorporate real-world tracking data to refine these corridors and ensure conservation measures are perfectly suited to the martens' actual ecological interactions.

Urban Vulnerability and Greenspace Accessibility Mapping

For my project, **"Urban Vulnerability and Greenspace Accessibility Mapping,"** I conducted a comprehensive spatial analysis of **Greater Manchester** to evaluate if current greenspace provision meets the needs of its residents. My primary objective was to benchmark accessibility against national standards and identify the most critical areas for new community greening interventions to improve public health and reduce urban inequality.

1. The Basics: Motivation and Conceptual Framework

Urbanization is often linked to deteriorating mental and physical health, including higher rates of depression and non-communicable diseases. However, I recognized that **greenspaces**—ranging from public parks to river corridors—act as vital buffers that improve air quality and provide space for recreation and physical activity.

To quantify "sufficiency," I adopted the **Accessible Natural Greenspace Standards (ANGSt)** set by Natural England as my benchmark. These standards provide specific size-to-distance thresholds:

- **2 hectares (ha)** within 300 meters of a home.
- **20 ha** within 2 km.
- **100 ha** within 5 km.
- **500 ha** within 10 km.

2. Data Acquisition: Types, Sources, and Rationales

I integrated five distinct categories of geospatial and demographic data to build my model:

Data Type	Source	Rationale for Use
Greenspace Geometry	**Ordnance Survey (OS)**	I used this to identify individual green space polygons across Greater Manchester.
Administrative Boundaries	**Office for National Statistics (ONS)**	I acquired Local Authority District (LAD) and Lower Super Output Area (LSOA) data to partition the analysis into meaningful urban units.
Population Density	**OpenPopGrid (Univ. of Southampton)**	I utilized high-resolution population rasters to accurately calculate how many individuals were within or outside access buffers.

Deprivation Ranks (IMD)	**UK Government National Statistics**	I included the **Index of Multiple Deprivation** to ensure the model prioritized interventions in the most disadvantaged communities.
Vegetation Health (NDVI)	**Sentinel-2 (ESA)**	I used satellite imagery to calculate the **Normalized Difference Vegetation Index**, allowing me to assess the actual "greenness" or quality of existing vegetation.

3. Technical Methodology and Key Decisions

My methodology involved a sophisticated multi-stage GIS workflow:

- **Public Access Filtering:** I made the critical decision to filter the OS dataset to include **only** publicly accessible areas. I excluded private gardens, business amenities, school grounds, and sports facilities. I also removed inland water to focus strictly on terrestrial greenspace.
- **Accessibility Modelling:** I created multi-ring buffers around sites that met the 2ha, 20ha, and 100ha thresholds. Using the **Select By Location** and **Dissolve** tools, I created consolidated polygons representing the total "service area" for each greenspace category.
- **Quality and Clustering Analysis:** I used **NDVI values** to measure vegetation health, where higher values indicated higher quality. I also performed a **Cluster and Outlier Analysis (Moran's I)** to determine if greenspaces were clustered in specific LSOAs or if there were significant provision "cold spots".
- **The Demand Map Logic:** To recommend new sites, I created a final **Demand Map** using a **Raster Calculator**. I aggregated four standardized demand rasters:
    1. Population density with no access.
    2. Zones of greenspace deficiency.
    3. LSOA deprivation ranks.
    4. Low NDVI values (indicating poor vegetation quality).

4. Quantitative Results and Findings

- **Provision Breakdown:** Out of **53,061 publicly accessible sites** in Greater Manchester (covering ~6% of the total area), only **692 sites (1.3%)** met the 2ha threshold. Only 41 sites were >20ha, and just 1 site was >100ha.
- **Accessibility Disparities:** I found that while **54% of the population** can access a 20ha site within 2km, only **13%** can access a 100ha site within 5km.
- **District Performance:** Trafford and Oldham demonstrated the highest percentage of residents with access to 2ha sites, while **Bolton and Bury** had the lowest.
- **Clustering Results:** My Moran's I analysis showed **no significant clustering** of greenspace in most LSOAs, highlighting a general deficiency in the urban core.

5. Conclusions and Recommendations

I concluded that current greenspace provision in Greater Manchester is insufficient, particularly regarding larger regional parks. Through my **Demand Map**, I identified high-priority zones where new interventions are most needed. I recommended that local planners target **brownfield sites** or unused public land within these high-demand zones to create new parks, which would simultaneously improve biodiversity and reduce health inequalities.

Limitations noted: My model used straight-line Euclidean distances for buffers. For future work, I would recommend using **network analysis** to measure distance along actual road paths, which would provide a more realistic representation of human movement.

Weighted Redistribution Algorithm

In this project, I implemented a **Weighted Redistribution Algorithm** in Python to solve a common problem in spatial analysis: **centroid bias** in passively geocoded data.

1. The Basics: The Problem of Passively Geocoded Data

I recognized that many spatial datasets, such as social media posts, are geocoded based on place names rather than precise GPS coordinates. This "passive" georeferencing results in heavy generalization; when plotted, all data points associated with a town name snap to that administrative area's **centroid**. This creates **false hotspots** in heatmaps that do not reflect where the activities actually occurred, making it impossible to identify real spatial patterns or correlations.

2. Data Acquisition: Types, Sources, and Rationales

To build and test my implementation of the algorithm proposed by Huck, Whyatt, and Coulton (2015), I integrated the following data:

Data Type	Source	Rationale for Use
Passively Geocoded Points	Level 3 Tweets (Royal Wedding)	I used a random sample of **1,023 tweets** geocoded to districts in Greater Manchester as a representative "ambiguous" dataset.
Administrative Boundaries	UK Government/ONS	I used vector files of Manchester's administrative districts to define the polygons within which points would be redistributed.
Weighting Surface	**Population Density**	I chose a population density raster of Greater Manchester. This provides a "sensible" surface, assuming tweets are more likely to originate from areas where more people live.

3. Technical Methodology and Key Decisions

My Python implementation followed a rigorous workflow to disaggregate the generalized points:

- **Pre-processing:** I used the rasterio library to handle the population density raster and ensured all vector data were converted to the **British National Grid** Coordinate Reference System (CRS) for accurate topological operations.
- **Spatial Indexing:** I utilized the .within() method to create a spatial index, allowing the script to efficiently identify which tweets belonged to which administrative polygon.
- **Generating Seed Locations:** For every tweet within a polygon, I used the numpy library to generate a series of **random points**. I used a while loop to verify that every generated point fell inside the administrative area.
- **Weight Selection:** I queried the population density weight for every random point. The point with the **highest weight** was selected as the new **"seed location"** for that tweet.
- **Modeling Ambiguity (The Likelihood Region):** Because I could not know the "true" original location, I avoided plotting a single point. Instead, I created a **likelihood region** (a disk) around the seed location using the skimage library's disk function.

4. Key Parameters and Mathematical Formulas

I carefully balanced the model using two user-defined variables:

- **Influence of Weight (**w=20**):** This value determines how many random points are generated. Increasing w forces points to cluster more tightly in high-density areas, while a lower w allows for more spatial variation.
- **Spatial Ambiguity (**s=0.01**):** This determines the radius of the likelihood region. I calculated the

  radius r using the equation: $r=\pi As$ *(where A is the area of the polygon)*.
- **Pixel Values (**v**):** Within the disk, I assigned pixel values to represent decaying likelihood from the

  center using: $v=1-r(seedx-cellx)2+(seedy-celly)2$.

5. Results and Conclusions

The final output was a **believable spatial arrangement**. By comparing the "before" and "after" plots, I demonstrated that:

- The data moved from arbitrary administrative centers to clusters in **high-population density regions**.
- The resulting map provided a much higher quality surface for heatmapping and subsequent spatial analysis than the original centroid-based map.

Reflections and Limitations: I concluded that while the algorithm significantly improves visualization, it remains an approximation. The points are not at their "actual" locations, and because the algorithm uses random point generation, the map results can vary slightly between runs. Nevertheless, this implementation remains a vital tool for correcting bias in large-scale geocoded datasets.

Passu Glacier Runoff Modelling

For my **Passu Glacier Runoff Modelling** project, I simulated glacial runoff dynamics to understand how temperature variations and physical glacier characteristics drive water release. This work was a key component of my **Environmental Monitoring and Modelling** module at the University of Manchester.

1. The Basics: Objective and Glacial Hydrology

Glacier hydrology is the study of how water moves through and is released from ice masses. My primary goal was to create a predictive model that could accurately estimate the runoff from the Passu Glacier based on air temperature data. Understanding these dynamics is critical because larger peak flows and faster water release from glaciers can lead to significant **upstream flooding** and impacts on downstream communities.

2. Data Acquisition: Types, Sources, and Rationales

To build a reliable simulation, I utilized several data types provided through the University of Manchester's monitoring datasets:

- **Air Temperature:** I used this as the primary independent variable to drive the melting process in the model.
- **Observed Runoff:** This historical data served as my "ground truth" to calibrate the model and evaluate the accuracy of my predictions.
- **Topographic and Environmental Parameters:** I integrated parameters like the **snowline elevation** and **lapse rate** to account for how temperature and snow coverage change with altitude.

3. Technical Methodology and Key Decisions

My modelling approach involved an iterative process of adjusting parameters to minimize the difference between my predicted runoff and the observed values:

- **The Degree Day Factor (DDF) Decision:** I tested the sensitivity of the model to the DDF for both ice and snow. I discovered that changing the DDF for ice had a negligible effect on predicted runoff, whereas **increasing the DDF for snow** was essential to bringing my predictions closer to reality.
- **Modelling Elevation Effects:** I made critical decisions regarding the **snowline elevation** and the **lapse rate**. I found that setting the snowline too high caused the model to underestimate runoff, while a snowline that was too low caused an overestimation. Furthermore, increasing the lapse rate led to a severe underestimation of the flow.
- **Drainage System Configuration:** I chose to model the sub-glacial environment as a **"conduit-dominated" system**. My rationale was that in such a system, water flows through the glacier more swiftly and efficiently, leading to faster and more concentrated flow reaching the terminus, which better reflects the high-energy environments of glaciers like Passu.

4. Quantitative Results and "Best Fit" Calibration

Through rigorous testing, I identified the specific mix of parameters that yielded the most accurate output:

- **DDF for Snow:** I achieved the best results with a value **close to 1**.
- **Snowline Elevation:** I determined the optimal elevation to be **2500 m**.
- **Lapse Rate:** I set this at **0.5°C** to maintain the most realistic temperature-to-altitude relationship.

My final model successfully captured the cyclical nature of the runoff, as shown in the **Passu Glacier runoff model plot** (Figure 4), which tracks the air temperature and predicted runoff over time with a **4-hour lag**.

5. Conclusions and Practical Implications

I concluded that the efficiency of water release is highly sensitive to the internal drainage structure of the glacier. By employing a conduit-dominated model, I demonstrated that larger peak flows are generated, which significantly increases the risk of flooding. This model proves that while temperature is the primary driver, the physical state of the glacier—specifically its snow coverage and internal plumbing—is what determines the ultimate intensity of the runoff. My findings provide a blueprint for understanding glacial responses to rising global temperatures and for managing associated hydrological risks.